Practical No. 9

Aim: Web Security with SSL/TLS: Configure and implement secure web communication using SSL/TLS protocols, including certificate management and secure session establishment.

Client.java

```
package clientserverpractical;
import java.util.*;
import java.io.*;
import java.net.*;

public class Client {
    public static void main(String[] args) {
        Socket socket=null;
        java.io.InputStreamReader insr=null;
        java.io.OutputStreamWriter oswr=null;
        java.io.BufferedReader br=null;
        java.io.BufferedWriter bw=null;
        try {
            socket =new Socket("localhost",8081);
            insr=new InputStreamReader(socket.getInputStream());
            oswr=new OutputStreamWriter(socket.getOutputStream());
            br= new BufferedReader(insr);
            bw= new BufferedWriter(oswr);
            Scanner sc=new Scanner(System.in);
            System.out.println("Send Message to server:");
            while(true) {
                String msgtosend=sc.next();
                bw.write(msgtosend);
                bw.newLine();
                bw.flush();
                if(msgtosend.equalsIgnoreCase("BYE")) {
                    break;
                }
            }
            bw.close();
        } catch(Exception e) {
            System.out.println("Error: "+e);
        }
    }
}
```

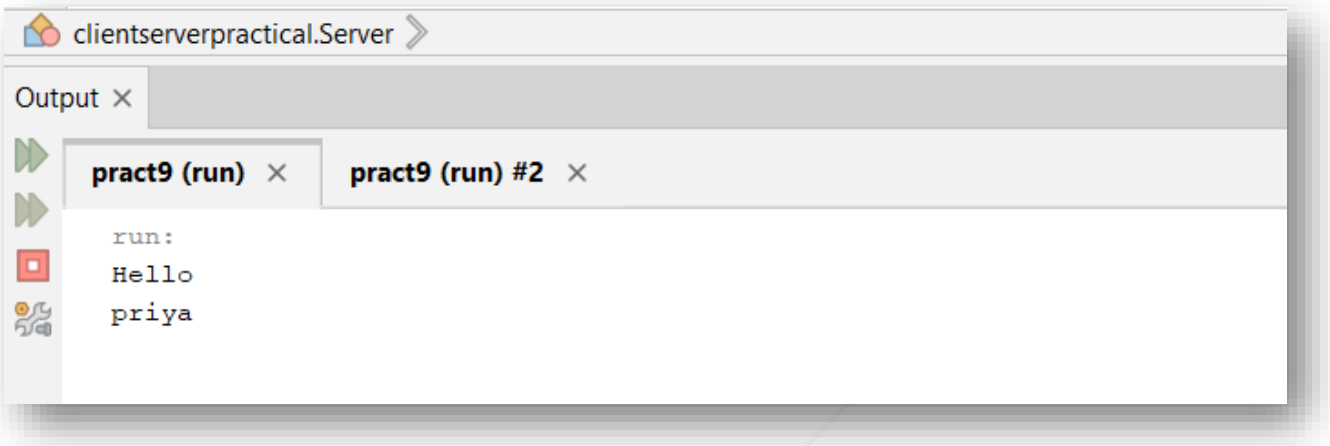
Server.java

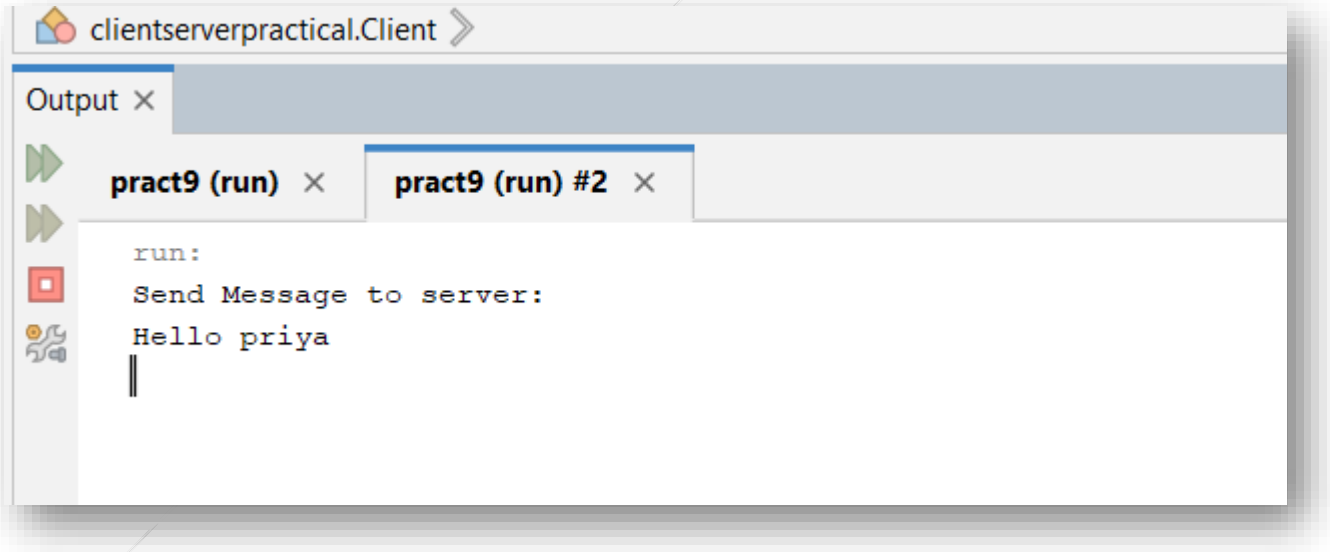
```
package clientserverpractical;
import java.io.*;
import java.net.*;

public class Server {
    public static void main(String[] args) throws IOException {
        Socket socket;
        ServerSocket serversocket=new ServerSocket(8081);
        java.io.InputStreamReader insr=null;
        java.io.OutputStreamWriter oswr=null;
        java.io.BufferedReader br=null;
        java.io.BufferedWriter bw=null;

        while(true) {
            try {
                socket=serversocket.accept();
                insr=new InputStreamReader(socket.getInputStream());
                oswr=new OutputStreamWriter(socket.getOutputStream());
                br= new BufferedReader(insr);
                bw= new BufferedWriter(oswr);
                while(true) {
                    String msgfromclient=br.readLine();
                    System.out.println(msgfromclient);
                    bw.write("Message Received");
                    bw.newLine();
                    bw.flush();
                    if(msgfromclient.equalsIgnoreCase("BYE")) {
                        break;
                    }
                }
                bw.close();
            } catch(Exception e) {
                System.out.println("Error: "+e);
            }
        }
    }
}
```

Output :


clientserverpractical.Server
Output
pract9 (run)
pract9 (run) #2
run:
Hello
priya


clientserverpractical.Client
Output
pract9 (run)
pract9 (run) #2
run:
Send Message to server:
Hello priya